

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Based Smart Immunization and Vaccination Management System

Abstract

Vaccination is one of the most effective methods of preventing infectious diseases and improving public health. However, many people still miss their vaccination schedules because they forget appointments, lose vaccination cards, or are unaware of the recommended immunization schedule. This issue mainly affects parents managing their children's vaccinations, adults requiring booster doses, pregnant women, and healthcare providers responsible for maintaining vaccination records. Missed vaccinations and incomplete records can increase the risk of vaccine-preventable diseases and reduce the overall effectiveness of immunization programs.

Most people currently rely on paper vaccination cards, hospital reminders, SMS alerts, or basic mobile applications to track their vaccination schedules. Although these methods provide some support, they have several limitations. Paper records can be lost or damaged, reminders are often generic, and many existing applications only provide notifications without analyzing vaccination history or recommending future vaccines. They also lack centralized lifetime vaccination records, family vaccination management, and intelligent decision support. Recent advancements in Artificial Intelligence have created opportunities to improve vaccination management through personalized and automated healthcare solutions.

The proposed AI-Based Smart Immunization and Vaccination Management System provides a secure and user-friendly platform for managing vaccination records throughout an individual's lifetime. The system stores digital vaccination records, sends automatic reminders, recommends upcoming vaccines using Artificial Intelligence based on age and vaccination history, detects missed doses, tracks vaccination progress, and provides chatbot support for common vaccination-related queries. It also supports family vaccination management, digital certificate generation, and downloadable reports, making vaccination tracking simple and efficient.

The scope of this project is limited to vaccination record management, AI-based vaccine recommendations, reminder generation, family vaccination tracking, chatbot assistance, and report generation. It does not include medical diagnosis, treatment recommendations, or hospital management. The proposed system aims to reduce missed vaccinations, improve record management, increase vaccination awareness, and provide a reliable digital platform that supports both users and healthcare providers in achieving better immunization coverage.

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References

1. This paper presents a digital vaccination management system with reminders and record tracking. It inspired our AI-based solution by adding personalized recommendations, smart reminders, chatbot support, and digital vaccination management.
“<https://ieeexplore.ieee.org/document/9445386>”
2. This paper focuses on improving vaccine distribution and immunization coverage through an optimized vaccination network. It inspired our project to develop an AI-based vaccination management system that enhances vaccine tracking, digital record management, and timely reminders for better immunization coverage.
“https://arxiv.org/abs/1907.13434?utm_source=”
3. **Immunization Agenda 2030 (IA2030)** is a global strategy developed by the World Health Organization (WHO) to improve vaccination coverage between 2021 and 2030. Its goal is to ensure that everyone, everywhere has equal access to life-saving vaccines. IA2030 encourages the use of digital technologies, effective immunization management, and timely vaccination to reduce preventable diseases and improve public health. This vision supports the development of smart vaccination systems like our proposed AI-Based Smart Immunization & Vaccination Management System.
“https://www.who.int/health-topics/vaccines-and-immunization#tab=tab_3”

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